The molecules in question are called chlorofluorocarbons (CFCs). Chemically, they're extremely inert, which means they're invulnerable—until they find themselves up in the ozone layer, where they're broken apart by ultraviolet light from the Sun. The chlorine atoms thus liberated attack and break down the protective ozone, letting more ultraviolet light reach the ground. This increased ultraviolet intensity ushers in a ghastly procession of potential consequences involving not just skin cancer and cataracts, but weakening of the human immune system and, most dangerous of all, possible harm to agriculture and to photosynthetic organisms at the base of the food chain on which most life on Earth depends.

Who discovered that CFCs posed a threat to the ozone layer? Was it the principal manufacturer, the DuPont Corporation, exercising corporate responsibility? Was it the Environmental Protection Agency protecting us? Was it the Department of Defense defending us? No, it was two ivory-tower, white-coated university scientists working on something else—Sherwood Rowland and Mario Molina of the University of California, Irvine. Not even an Ivy League university. No one instructed them to look for dangers to the environment. They were pursuing fundamental research. They were scientists following their own interests. Their names should be known to every schoolchild.

In their original calculations, Rowland and Molina used rate constants of chemical reactions involving chlorine and other halogens that had been measured in part with NASA support. Why NASA? Because Venus has chlorine and fluorine molecules in its atmosphere, and planetary aeronomers had wanted to understand what's happening there.

Confirming theoretical work on the role of CFCs in ozone depletion was soon done by a group led by Michael McElroy at Harvard. How is it they had all these branching networks of halogen chemical kinetics in their computer ready to go? Because they were working on the chlorine and fluorine chemistry of the atmosphere of Venus. Venus helped make and helped confirm the discovery that the Earth's ozone layer is in danger. An entirely unexpected connection was found between the atmospheric photochemistries of